SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION   COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)

Assessment Tool Weightage: 20%

Code of Conduct

I G. Gopi Chand (192424384) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Gopi chand.

1. Smart Surveillance System Selection

Constraints: latency must be ≤ 100 ms (otherwise the system fails); accuracy must be ≥ 85% (otherwise unacceptable); hardware supports only moderate computation.

Model	Accuracy	Latency	Meets Accuracy ≥85%?	Meets Latency ≤100ms?
Viola-Jones	78%	Low	No (78% < 85%)	Yes
YOLO	92%	Moderate	Yes	Yes
Deep Learning	95%	High	Yes	Likely No (high latency)

Comparison: Viola-Jones satisfies the latency condition easily but fails the accuracy threshold outright (78% < 85%), disqualifying it regardless of its speed. Deep Learning satisfies accuracy comfortably (95%) but its high latency puts it at risk of breaching the 100 ms ceiling, and its computational demand exceeds the 'moderate computation' hardware limit. YOLO meets the accuracy requirement (92% ≥ 85%) with moderate latency that fits within the 100 ms budget and moderate computational load matching the available hardware.

Evaluation: Applying the constraints as hard filters (not preferences) eliminates Viola-Jones on accuracy and Deep Learning on latency/hardware fit, leaving YOLO as the only model that simultaneously clears every stated condition.

Justification: YOLO is selected because it is the only option satisfying all three constraints together — accuracy ≥85%, latency within the 100 ms limit, and computational load compatible with moderate hardware — making it the sole logically valid choice, not merely the best trade-off.

2. Autonomous Vehicle Detection Logic

Constraints: accuracy must be > 90% (safety priority); response time must be < 50 ms.

Method	Accuracy	Response Time	Meets Accuracy >90%?	Meets Time <50ms?
Method A	88%	30 ms	No (88% ≤ 90%)	Yes
Method B	93%	70 ms	Yes	No (70 ms ≥ 50 ms)
Method C	91%	45 ms	Yes	Yes

Comparison: Method A is fast (30 ms) but fails the safety-critical accuracy threshold (88% is below the required >90%), which is disqualifying for a vehicle detection system regardless of speed. Method B clears the accuracy bar comfortably (93%) but its 70 ms response time exceeds the 50 ms limit, introducing an unacceptable reaction delay for a moving vehicle. Method C satisfies both conditions simultaneously (91% accuracy, 45 ms response).

Evaluation: Because safety is explicitly prioritized, accuracy below 90% cannot be traded for speed — this immediately removes Method A. Between the two accuracy-qualified options, Method B's latency violates the hard response-time constraint, leaving Method C as the only fully compliant method.

Justification: Method C is chosen because it is the only method meeting both the safety-driven accuracy requirement and the real-time response-time requirement; in a safety-critical autonomous driving context, constraint compliance takes precedence over marginal accuracy gains (as in Method B) or marginal speed gains (as in Method A).

3. Background Modeling Evaluation

Requirement: the system must maintain ≥ 80% accuracy across all scenarios. Observed GMM performance: Static scenes 90% accuracy; Dynamic scenes 65% accuracy.

Scenario	GMM Accuracy	Meets ≥80% Requirement?
Static	90%	Yes
Dynamic	65%	No (falls 15 points short)

Comparison: GMM comfortably exceeds the 80% requirement in static scenes (90%) but falls well short of it in dynamic scenes (65%), a 25-point gap between the two scenario types. This inconsistency shows GMM's single global adaptation model cannot handle both scenario types equally.

Evaluation: Since the requirement is 'across all scenarios' (i.e., the minimum must hold in every case, not on average), GMM alone does not meet the requirement — one non-compliant scenario is enough to fail the overall condition, even though the other scenario passes comfortably.

Justification: GMM alone should not be deployed as-is. This is a case where no single-technique option satisfies all constraints, so the logical decision is to modify the approach rather than force-fit an unsuitable model: augment GMM with adaptive learning rates, shadow suppression, or fuse it with a complementary technique (e.g., optical-flow-based motion detection) specifically to raise dynamic-scene accuracy above 80% while preserving its strong static-scene performance.

4. Depth Estimation Strategy

Constraints: only one camera is available; moderate depth accuracy is acceptable.

Option	**Accuracy**	**Camera Requirement**	**Feasible with 1 camera?**
Stereo Vision	High	Two cameras	No — hardware requirement not met
Structure from Motion (SfM)	Moderate	Single camera	Yes

Comparison: Stereo Vision offers higher depth accuracy but structurally depends on a second camera for disparity computation — a hardware requirement the system cannot satisfy under the given single-camera constraint, regardless of its accuracy advantage. Structure from Motion reconstructs depth from a sequence of frames captured by a single moving camera, and its moderate accuracy is explicitly acceptable per the stated requirement.

Evaluation: The camera-count constraint acts as a hard feasibility filter, not a performance trade-off: an option that needs hardware the system does not have cannot be implemented at all, no matter how accurate it is. This immediately rules out Stereo Vision.

Justification: Structure from Motion is the only feasible option given the single-camera constraint, and since the requirement explicitly accepts moderate accuracy, SfM's performance level is sufficient — it is both technically implementable and requirement-compliant, whereas Stereo Vision is not implementable at all under the given hardware limitation.

5. Mobile Face Detection Trade-off

Constraints: real-time operation is required; accuracy must be $\geq 85\%$.

Approach	Accuracy	Speed	Meets Accuracy ≥85%?	Meets Real-time Requirement?
Viola-Jones	80%	Fast	No (80% < 85%)	Yes
Deep Learning	95%	Slow	Yes	No (slow, not real-time)

Comparison: Viola-Jones is fast enough for real-time mobile use but falls short of the minimum accuracy requirement (80% < 85%), so speed alone does not make it compliant. Deep Learning clears the accuracy bar by a wide margin (95%) but its slow inference conflicts directly with the real-time operation requirement, which on typical mobile hardware makes it impractical to deploy as-is.

Evaluation: Neither option satisfies both constraints simultaneously in their base form — this is a genuine case where no listed option is fully compliant, requiring a reasoned compromise rather than a simple pick between the two.

Justification: Since accuracy below the 85% threshold is a hard failure for face detection (false negatives are unacceptable) while a modest, optimized deep model (e.g., a compressed/quantized or lightweight CNN-based detector) can approach real-time speed on mobile hardware, Deep Learning — with mobile-specific optimization (model compression, quantization, or a lightweight architecture) — is the more defensible choice, since accuracy can only be recovered post-hoc for Viola-Jones through algorithmic changes that fundamentally alter it, whereas speed can be improved for Deep Learning through standard mobile-deployment optimization without sacrificing its accuracy advantage.